Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Agentic AI Frameworks, GenAI, Retrieval-Augmented Generation (RAG), Prompt Engineering, Model Fine-Tuning, Dimensional Data Modeling, Star/Snowflake Schemas, Slowly Changing Dimensions (SCD Type 1/2), MLOps, Data Governance, Anomaly Detection, Predictive Modeling, Time-Series Forecasting (SARIMA), Root Cause Analysis, CI/CD

Programming Languages & Tools: Python, PyTorch, TensorFlow, Hugging Face Transformers, Scikit-Learn, LangChain, AutoGen, Azure Cloud (ADF, Synapse, SQL, Databricks), AWS (S3, Redshift), Snowflake, Apache Spark, PySpark, Docker, Kubernetes, Power BI, Tableau, Airflow

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

AI / ML & Data Engineering Associate — **May 2022 – Present**

- **Architected Agentic AI and GenAI Frameworks** Designed and deployed conversational AI and automated QA testing frameworks (SpectrumIQ) cutting manual testing turnaround by 50% while applying prompt engineering, automated profiling, and regex classification for complex data pipelines.
- **Implemented Predictive Machine Learning and Anomaly Detection** Developed proactive anomaly detection and predictive modeling systems using XGBoost, Random Forests, Scikit-Learn, and SARIMA with 95% accuracy, mitigating ETL discrepancies and optimizing resource allocation by 20%.
- **Optimized High-Throughput Data Infrastructure and Pipelines** Engineered high-performance SQL stored procedures, dimensional data modeling (Star/Snowflake schemas), and Slowly Changing Dimensions (SCD Type 1/2) for scalable processing of structured and unstructured datasets.
- **Automated Data Governance and Quality Auditing** Built automated Data Quality Audit frameworks equipped with validation firewalls and data governance protocols, reducing operational troubleshooting turnaround by over 80% and ensuring 99.9% data availability.
- **Operationalized MLOps and Cloud-Scale Analytics** Managed end-to-end cloud deployments across Azure (ADF, Synapse, Databricks, SQL) and AWS (S3, Redshift), leveraging Apache Spark/PySpark, Airflow batch monitoring, and containerized architectures for production-ready reliability.

Master of Science in Information Systems & Data Science — **May 2022 – Present**

- **Mastered Advanced Statistical and Machine Learning Methodologies** Completed rigorous coursework in information systems and data science at the University of Cincinnati, achieving a 3.95/4.0 GPA with specialized focus on predictive algorithms and data analytics.
- **Applied Deep Learning and Quantitative Problem Solving** Executed applied machine learning research projects emphasizing data modeling, computational efficiency, and advanced statistical analysis for large-scale enterprise datasets.
- **Bridged Enterprise Software Engineering with Data Science** Integrated theoretical foundations of data structures, database management systems, and enterprise architecture into practical software development lifecycles.

Bachelor of Engineering in Petrochemical — **May 2022 – Present**

- **Built Strong Analytical and Mathematical Foundations** Completed core engineering curriculum at Pune University MIT, focusing on complex mathematical modeling, numerical optimization, and system reliability.
- **Executed Complex Process Automation and Optimization Projects** Analyzed large multivariate data streams to troubleshoot operational bottlenecks and improve system-wide throughput and efficiency.
- **Collaborated in Multidisciplinary Technical Environments** Partnered with cross-functional teams to design, test, and maintain robust operational workflows under strict safety and compliance standards.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture **Pune, India**

Test Engineering Analyst **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.